

ZINDER, NigerWMO: **610900**Lat: **13.779N**Lon: **8.984E**Elev: **462**StdP: **95.90**Time Zone: **1.00 (E01)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	14.0	15.1	-8.2	2.0	27.8	-8.2	2.0	27.9	9.3	22.1	8.4	22.1	2.6	40	0.433

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	12.5	41.8	19.2	41.0	19.4	40.2	19.5	25.6	33.0	25.1	32.4	24.7	32.0	3.3	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.9	19.9	27.9	23.4	19.3	27.8	23.0	18.8	27.7	81.3	33.1	79.4	32.4	77.9	32.0	30.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.3	7.5	6.9	11.2	43.4	1.2	0.7	10.3	43.9	9.6	44.3	8.9	44.7	8.1	45.2		
(5)			4.4	27.6	0.5	1.1	4.0	28.4	3.8	29.0	3.5	29.6	3.2	30.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	29.0	22.4	26.4	30.1	33.4	34.0	32.4	29.7	28.1	29.7	30.8	27.2	23.4
	DBStd	4.12	2.77	3.14	2.44	1.85	1.70	2.12	2.12	1.79	1.81	1.35	2.08	2.31
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	1	1	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6926	386	459	622	701	746	673	611	561	591	644	516	417
	CDD18.3	3885	128	226	364	451	487	423	353	303	341	386	266	158
	CDH23.3	53173	1296	2766	5232	7304	7929	6304	4501	3339	4299	5419	3223	1560
CDH26.7	31576	543	1511	3239	5007	5492	4014	2371	1496	2283	3238	1736	645	
Wind	WSAvg	3.4	3.8	3.6	3.7	3.3	3.4	4.0	4.1	3.2	2.6	2.7	3.1	3.6
Precipitation	PrecAvg	411	0	0	1	1	17	41	124	163	58	9	0	0
	PrecMax	655	1	5	19	17	90	100	291	350	172	160	4	0
	PrecMin	219	0	0	0	0	0	0	19	0	8	0	0	0
	PrecStd	111	0	1	3	3	19	28	57	70	37	24	1	0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.0	39.2	41.0	42.9	42.9	41.2	38.9	36.0	38.9	39.6	37.9	35.2
		MCWB	15.8	17.1	17.5	18.3	19.8	21.9	23.0	24.2	20.4	18.8	17.3	15.9
	2%	DB	34.0	37.8	39.9	41.9	41.9	40.2	37.1	34.9	37.5	38.8	36.8	33.6
		MCWB	14.7	16.4	17.1	18.6	19.5	22.1	23.5	24.4	22.1	18.9	16.7	15.1
	5%	DB	31.9	36.1	38.9	41.0	41.1	39.2	36.0	33.9	36.2	37.9	35.4	32.0
		MCWB	13.9	15.5	16.7	18.6	19.8	22.5	23.6	24.3	23.1	19.0	16.2	14.2
	10%	DB	30.0	34.4	37.8	40.1	40.2	38.1	34.8	32.8	35.1	37.0	34.2	30.4
		MCWB	13.2	14.9	16.1	18.7	20.1	22.7	23.7	24.1	23.6	19.2	15.9	13.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.2	18.2	20.9	24.0	25.1	25.8	26.1	26.0	26.2	25.2	20.2	16.9
		MCDB	34.0	36.8	35.2	37.0	35.8	35.8	33.7	32.6	33.1	32.7	33.8	32.2
	2%	WB	15.3	17.2	19.0	23.0	24.2	25.0	25.1	25.3	25.5	24.3	18.1	15.7
		MCDB	32.3	35.7	36.2	35.8	35.3	34.8	32.3	31.7	32.4	32.3	34.2	32.3
	5%	WB	14.2	16.1	17.5	22.1	23.7	24.5	24.7	24.9	25.0	23.6	17.1	14.8
		MCDB	31.0	34.6	36.7	34.8	34.9	34.1	31.8	31.2	31.9	31.7	33.8	30.7
	10%	WB	13.3	15.2	16.5	21.3	23.1	24.0	24.3	24.6	24.6	22.8	16.2	13.8
		MCDB	29.6	33.4	36.2	34.7	34.3	33.6	31.4	30.6	31.4	30.8	32.8	29.4
Mean Daily Temperature Range	5% DB	MDBR	13.6	14.2	14.0	13.7	12.5	11.3	9.9	8.9	10.8	12.9	13.9	13.6
		MCDBR	16.2	15.9	15.6	14.3	13.6	12.3	11.2	10.2	12.2	14.0	15.1	15.4
		MCWBR	6.6	6.0	5.8	5.8	5.8	3.9	2.9	3.0	3.8	4.8	5.2	6.1
	5% WB	MCDBR	15.9	15.2	14.6	12.8	12.0	11.3	10.0	9.4	10.8	11.9	14.7	15.0
		MCWBR	6.9	6.3	6.2	6.2	5.0	3.8	3.2	3.3	3.7	4.6	5.7	6.3
Clear-Sky Solar Irradiance	taub	0.557	0.617	0.670	0.722	0.709	0.700	0.649	0.613	0.618	0.653	0.556	0.546	
	taud	1.710	1.567	1.478	1.425	1.465	1.528	1.676	1.750	1.721	1.584	1.684	1.721	
	Ebn at Noon	740	717	694	658	654	652	689	724	721	680	733	734	
	Edh at Noon	231	277	310	326	307	285	247	232	239	269	235	223	
All-Sky Solar Radiation	RadAvg	5.66	6.12	6.49	6.66	6.61	6.31	6.04	5.94	6.32	6.13	5.79	5.47	
	RadStd	0.19	0.16	0.27	0.22	0.23	0.25	0.17	0.20	0.15	0.19	0.16	0.20	